

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. (EC) Semester - III Examination, 2010
EE - 220 Electrical Machines and Power Circuits

Date: 01-12-2010, Wednesday

Time: 10.30a.m to 1.30p.m

Maximum Marks: 70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicate *full* marks.
3. Make suitable assumptions and neat figures if required.

SECTION- I

Q1 (a) Attempt all questions.

[10]

- (1) Give Significance of back e.m.f in case of d.c. motor.
- (2) Why D.C series motor never started on no load?
- (3) Compare D.C motor and D.C generator.
- (4) Compare lap winding and wave winding.
- (5) How single-phase induction motor can be self started?

Q2 (a) List out the parts of D.C machine and explain function of each part.

[07]

- (b) A 4 pole, lap wound, d.c. generator has a useful flux of 0.07 wb per pole. Calculate the generated e.m.f. when it is rotated a speed of 900 r.p.m. with the help of prime mover. Armature consists of 440 number of conductors. Also calculate the generated e.m.f. if lap wound armature is replaced by wave wound armature.

[06]

OR

Q2 (a) Explain how rotating magnetic field is produced in 3 phase induction motor.

[07]

- (b) The rotor resistance and standstill reactance of a 3 phase induction motor are respectively 0.015 ohm 0.09 ohm per phase. At normal voltage, the full-load slip is 3%. Estimate the percentage reduction in stator voltage to develop full-load torque at half full-load speed. Also calculate the power factor.

[06]

Q3 Attempt any two .

[12]

- (a) Write a technical note on losses of D.C machine.
- (b) Explain the characteristics of series shunt motor.
- (c) Explain capacitor Starts- and- Run induction motor.

FOR REFERENCE



P.T.O...

SECTION - II

Q4 () Attempt all questions.

- (1) Why Alternators are rated in kVA rather than in kW? [01]
- (2) Explain the relation between Torque and Slip. [03]
- (3) Explain Pitch factor, Distribution factor and calculate the pitch factor of the following winding: 36 stator slots, 4 poles, coil span 1-8. [03]
- (4) Draw the vector diagrams for loaded alternators. [03]

Q5 (a) Explain the synchronizing of Three phase alternators. [07]

- (b) A 100 kw (output), 3300V, 50Hz, 3 phase star connected induction motor has a synchronous speed of 500 RPM. The full load slip is 1.8% and FL power factor 0.85. Stator copper loss = 2440W. Iron loss = 3500W. Rotational losses = 1200W. Calculate (i) the rotor copper loss (ii) the line current (iii) the full load efficiency. [06]

OR

Q5 (a) Enlist the methods for starting squirrel cage motors. Explain the star delta starter. [07]

- (b) A 60-kVA, 220 V, 50 Hz, 1-phase alternator has effective armature resistance of 0.016 ohm and an armature leakage reactance of 0.07 ohm. Compute the voltage induced in the armature when the alternator is delivering rated current at a load power factor of (a) unity (b) 0.7 lagging (c) 0.7 leading. [06]

Q6 Attempt any two. [12]

- (a) Explain the classification of sub-stations.
- (b) List out the types of Tariffs and explain each.
- (c) Enlist the methods for improving the power factor and explain any one method detail.
